



Is the model wrong randomly, or wrong systematically?

Use Case 1: Why Does Our Salary Model Make Mistakes?

Regression Error Analysis — Employee Salary Prediction

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Advanced Topics in Data Science · Reichman University

How We Answered the Question

Three layers — narrowing from how much, to where, to why



Global Metrics

How wrong on average?

MAE & RMSE across the test set



Residual Structure

Is there a pattern?

Shape and bias hidden in the errors



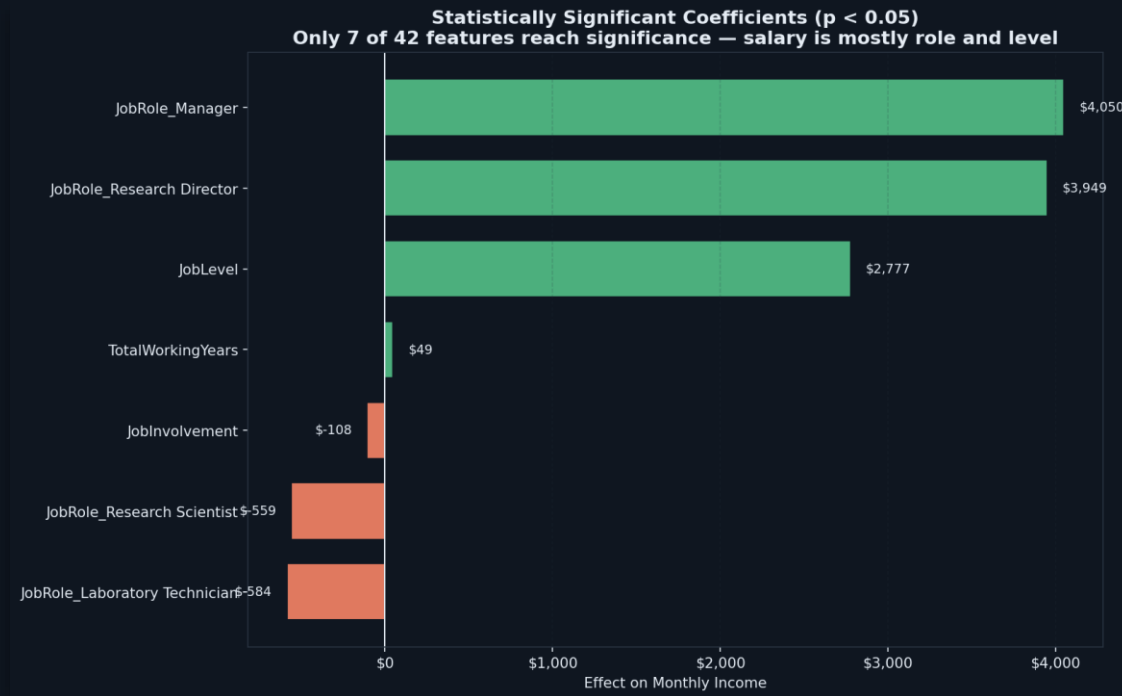
Segment Errors

Which groups fail?

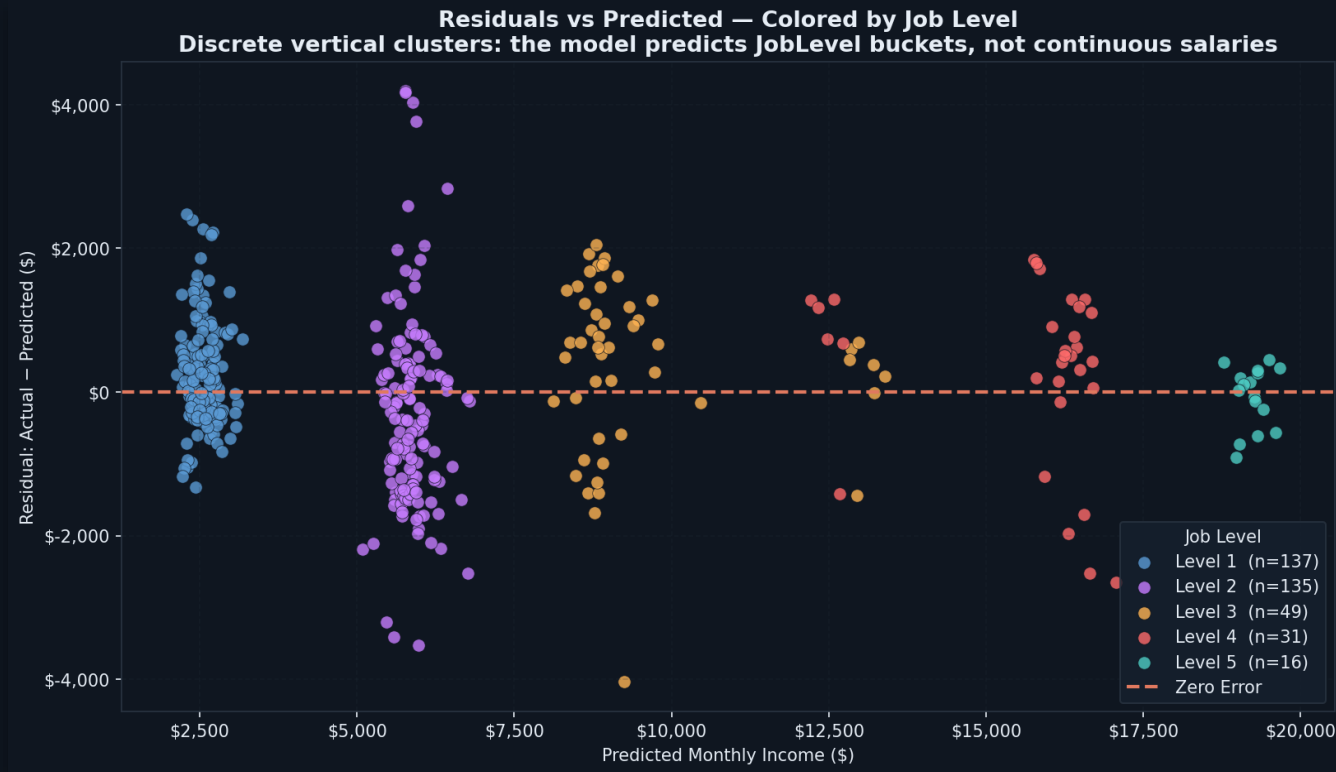
Error broken down by job level & role

Which Coefficients Are Statistically Significant?

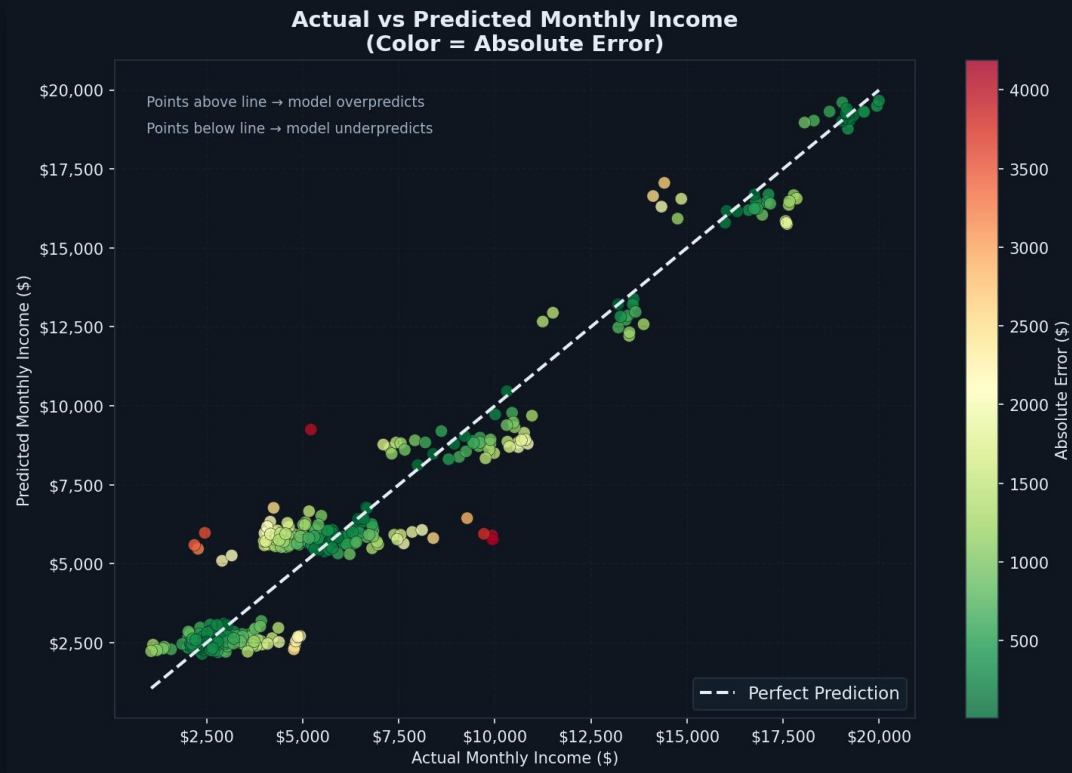
- $R^2 = 0.94$ on the training set
- Only 7 features are significant ($p < 0.05$)
- Job level and role dominate everything else
- Each job level adds ~\$2,777 / month



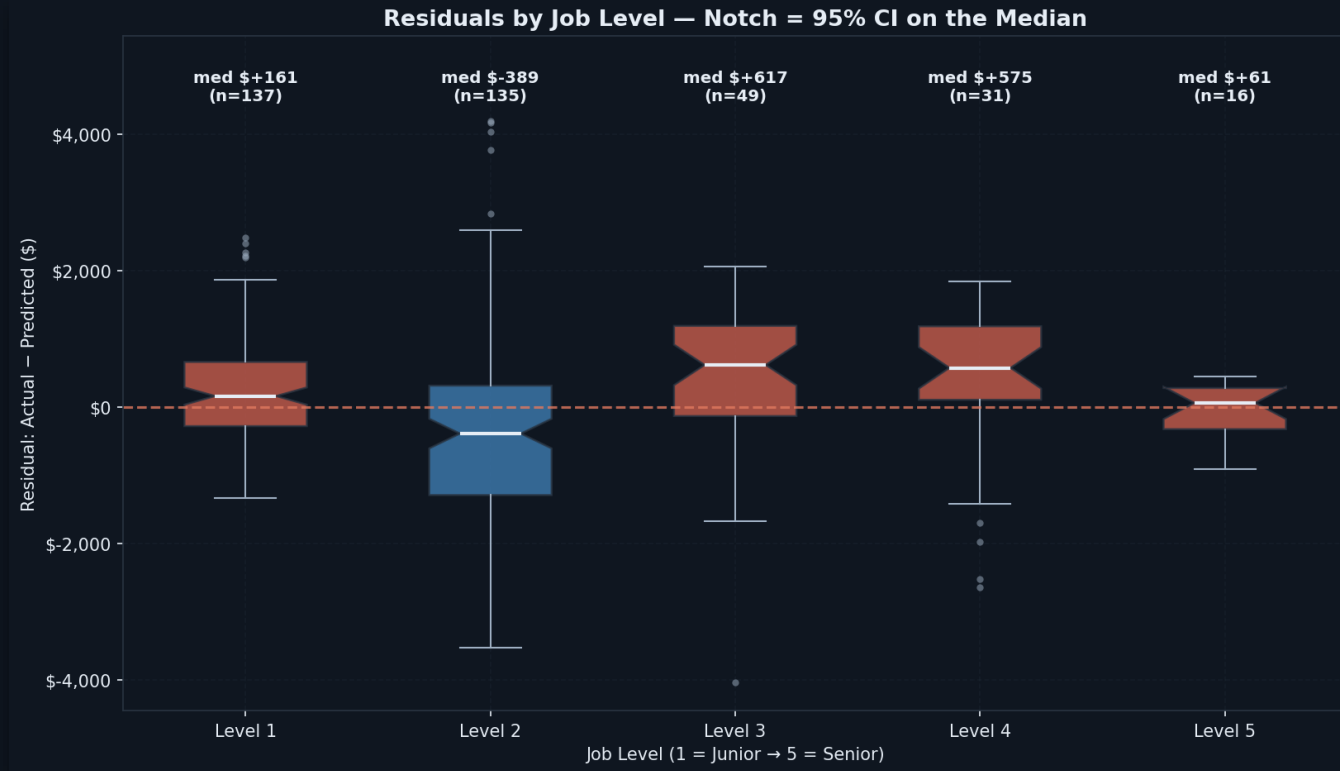
Are There Visible Patterns in the Residuals?



Does the Model Perform Equally Well Across Salary Ranges?

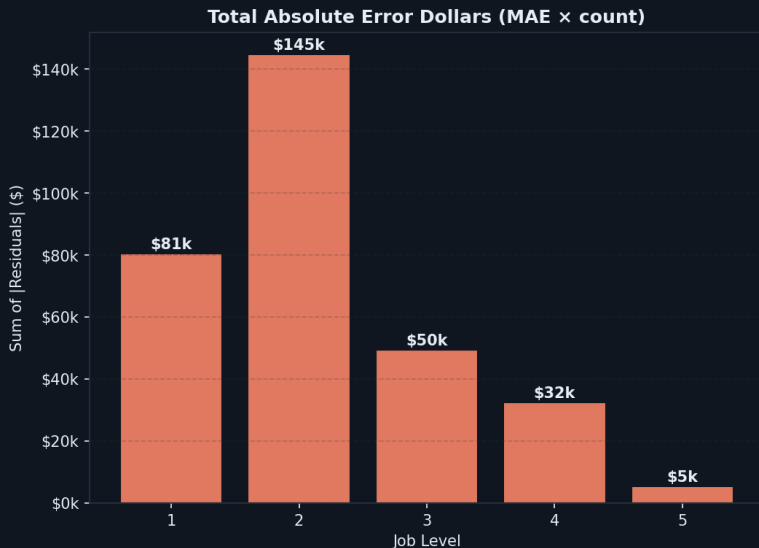
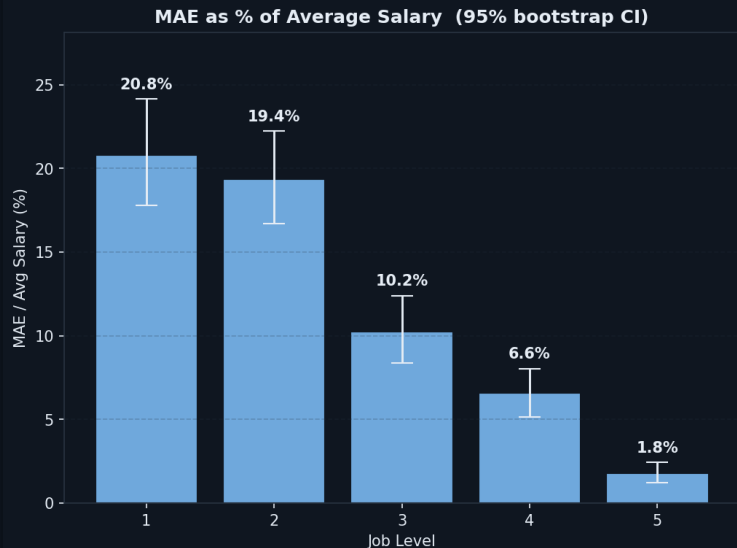


Is the Model Systematically Over- or Under-Predicting?



Are High Salaries Harder to Predict?

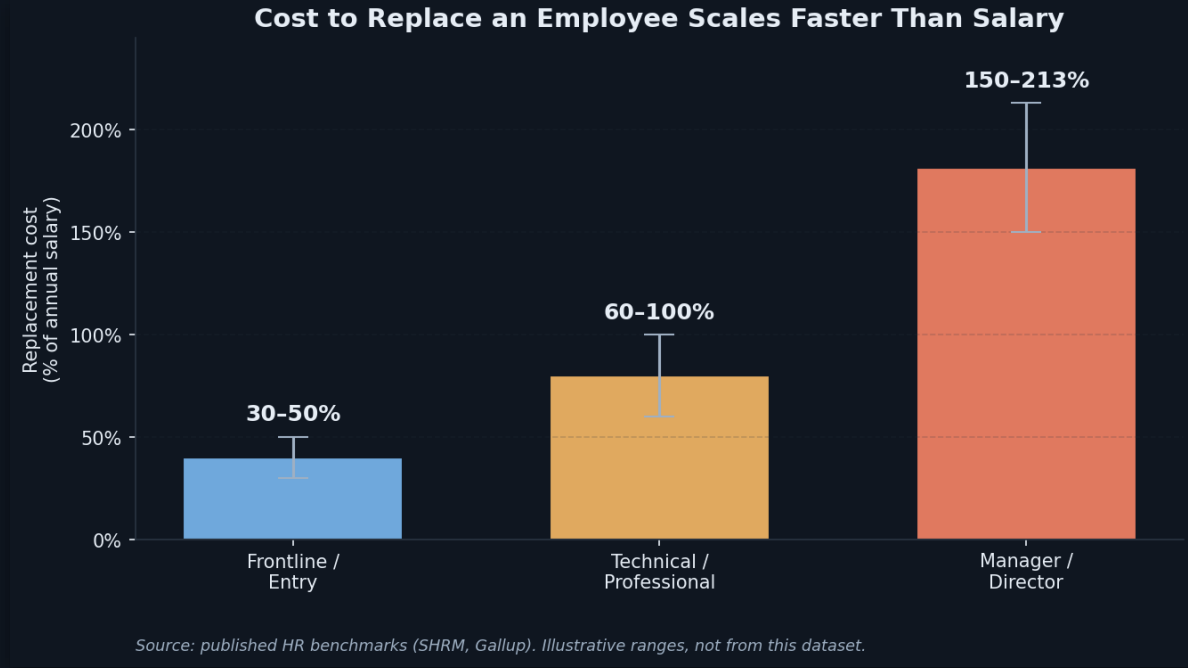
Where Errors Hurt Most: Two Lenses on the Same Errors



- Lens switch: dollars → % of salary
- Juniors: ~21% off · Seniors: ~2% off
- Bootstrap 95% CIs confirm the gap is real
- Level 2 carries the most total error dollars

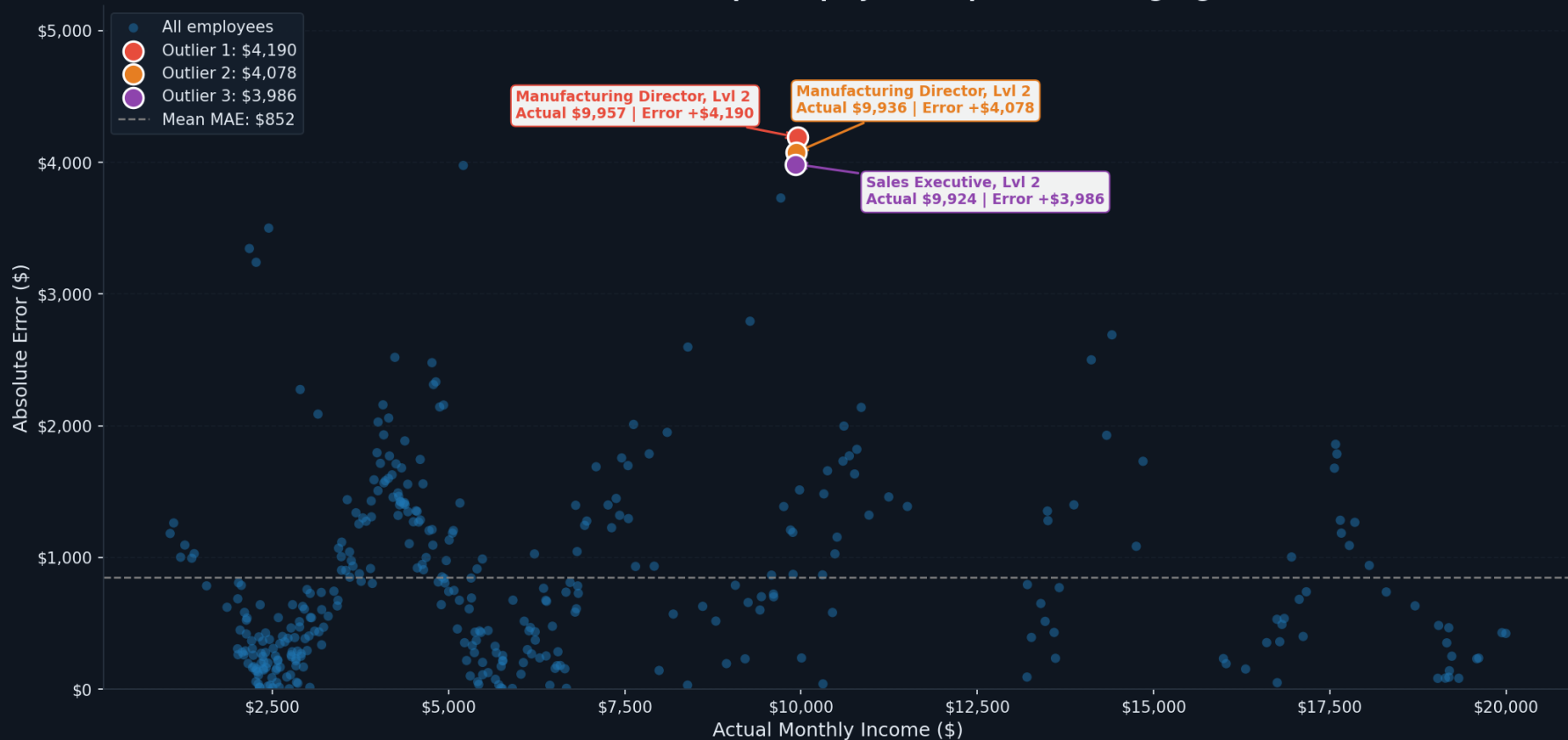
Where Would Predictions Be Dangerous?

- Replacement cost: ~40% of salary (frontline) → ~200% (managers / directors)
- Attrition runs the other way — higher in lower-salary groups (last semester)
- A wrong Director benchmark risks a ~\$300k+



Why Did These Specific Predictions Fail?

Absolute Prediction Error per Employee — Top 3 Outliers Highlighted



Should HR Deploy This Model?



Wrong systematically, not randomly



Not reliable for setting or benchmarking individual salaries



Valuable as a diagnostic — errors flag where pay breaks from job-level structure

From the rubric — what comes next

- *What features are missing?* Within-level signals — performance, bonus history, time-in-role — plus an attrition flag to link pay gaps to who leaves
- *Which additional data would improve performance?* Comparable-company / market salary data — for calibration and benchmarking, not blind training
- *What would you improve?* Start with a proper train / validation / test split — this model was fit, not selected



References (replacement-cost benchmarks)

[“The Cost of Replacing an Employee and the Role of Financial Wellness” \(Enrich\)](#)

[“The Cost of Losing Employees” \(Payscale\)](#)

[“The True Cost of Employee Turnover: Hidden Expenses That Hurt Your Business”
\(clearly rated\)](#)

[“This Fixable Problem Costs U.S. Businesses \\$1 Trillion” \(Gallup\)](#)

[Jupyter Notebook](#)